

STATEMENT OF PURPOSE

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Cultivated with algorithmic thinking from an early stage—particularly the aptitude for “divide and conquer” paradigms and structural optimization—my initial foray into Electrical Engineering provided me with a robust grounding in traditional physical operations. However, I soon realized that to surmount the challenges of the future grid, the system cannot solely rely on mechanistic protocols or the exorbitant expansion of hardware infrastructure. My academic fervor was truly ignited when I delved into the core operational principles and began perceiving the power grid through the lens of Computer Science. Transcending the physical facade of conductors and transformers, I was astounded to recognize the national power system as fundamentally a colossal mathematical graph, wherein the set of vertices (load nodes) and edges (transmission lines) are directly configured by impedance weights (R , X). Therein, all energy flows are strictly constrained by rigorous systems of equations, rendering the power grid one of the world’s most intricate, large-scale optimization problems. At this intersection, I apprehended the decisive role of computational logic in orchestrating practical power systems. Rather than depending exclusively on conventional solutions, sophisticated codebases and computational models facilitate the simulation, control, and maximal exploitation of the safe operational limits of existing physical equipment. This paradigm shift has crystallized my research objective: to leverage a computer science foundation and programming proficiency to solve system control problems, thereby addressing the technical exigencies of the Smart Grid in the era of renewable energy integration.

During my tenure as a researcher at the PGRE Laboratory under the mentorship of Assoc. Prof. Nguyen Duc Tuyen, coding simulations for the Unit Commitment problem profoundly shaped my methodological approach: utilizing algorithmic structural optimization as the cornerstone to decipher the complexities of power systems. Commencing with Linear Programming (LP), I escalated the model to Mixed-Integer Linear Programming (MILP) by establishing discrete variables to manage the operational states of conventional generators and the charge/discharge cycles of Battery Energy Storage Systems (BESS). Confronted with the non-convex solution space and the NP-hard complexity inherent in the AC power flow problem, I implemented SOCP convex relaxation techniques. This mathematical restructuring is intrinsically analogous to the logic of algorithmic complexity reduction: mapping a convoluted nonlinear system of equations onto a convex search space. This transformation accelerates algorithmic convergence, effectively mitigates power curtailment, and rigorously upholds voltage and reactive power constraints.

Nevertheless, as the scale of the grid graph expands, the combinatorial explosion of the variable space induces a computational bottleneck, causing centralized solvers to hit execution time limits. To address system resilience post-faults, I architected a distributed

optimization framework through my thesis research—which culminated in a 1st Author publication at ICGEA—by decomposing the IEEE 123-bus system into 4 autonomous Microgrids capable of P2P trading. This framework integrates the Alternating Direction Method of Multipliers (ADMM) with Model Predictive Control (MPC)—triggered instantaneously upon the occurrence of faults with a 5-hour horizon. During the implementation of parallel processing, I directly confronted the intrinsic divergence phenomenon of ADMM. By developing an adaptive penalty tuning mechanism, I successfully enforced convergence, drastically compressing the required iterations from an average of 150 to a mere 25-30 iterations, thereby remarkably accelerating computational speed. Although the SOCP-ADMM-MPC architecture exhibits exceptional responsiveness to anomalies within a short receding window, extending the prediction framework beyond this 5-hour horizon precipitates an exponential surge in the online computational burden, intrinsically bounding the system’s capacity for long-term strategic routing.

Despite having forged a robust technical foundation, I soon recognized a more formidable challenge: the objective limitations of an energy system in its nascent stage of transition. The power grid in Vietnam is currently designed and operated under a centralized, top-down dispatch architecture. In a context where physical infrastructure development lags behind the rapid penetration of renewable energy, the system is compelled to unconditionally prioritize operational security over solving complex optimization paradigms. This invariably leads to the prevalent, palliative measure of passive power curtailment. Furthermore, within the current milieu, avant-garde concepts such as Demand Response or Microgrids encounter substantial impediments due to the dearth of highly algorithm-compatible research frameworks required for empirical validation. Consequently, to holistically engineer core technologies for the Smart Grid, I must immerse myself in a vanguard academic ecosystem—one equipped with state-of-the-art laboratory facilities and meticulously digitized Wholesale Electricity Markets and retail models that are empirically operational.

The paradigm shift towards a decentralized electricity market is an inexorable global trajectory in the renewable energy epoch. To prepare for this transition, the United States stands as the preeminent destination. Although confronting similar infrastructural hurdles such as large-scale curtailment, the true prowess of the U.S. power grid resides in the maturity of its Wholesale Electricity Markets, orchestrated by Independent System Operators and Regional Transmission Organizations (ISO/RTO). It is precisely this transparent and intricate market environment, underpinned by Locational Marginal Pricing (LMP) mechanisms, that provides the flawless crucible for validating autonomous dispatch algorithms. Cognizant of the computational bottlenecks of current MPC models, my short-term research objective in a U.S. graduate program is to architect a hybrid control framework synthesizing MPC with Machine Learning and Reinforcement Learning (ML/RL) methodologies. By migrating the colossal computational workload to an offline training phase to exploit the value function approximation capabilities of AI, I aim to unshackle the system’s long-term horizon

limits while retaining MPC as a “safety filter” to preserve inviolable physical constraints. This research ambitiously seeks to formulate a comprehensive decision-making mechanism for energy trading and economic dispatch within Distributed Energy Resource Management Systems (DERMS) deeply integrated into the LMP market architecture. In the long term, I aspire to become a pivotal expert, repatriating core technologies and avant-garde electricity market design philosophies to Vietnam. My vision is to catalyze the transformation of my homeland’s grid: evolving from passively “curtailing renewables for security” to proactively “harnessing algorithmic supremacy to dictate stability and optimize economic utility.”

The [Program Name] program at [University Name] constitutes the perfect crucible for this vision. I am profoundly captivated by the research trajectory of Professor [Professor Name] concerning [Research Field]. The prospect of joining the [Lab/Research Center Name] would furnish an ideal ecosystem to empirically test distributed control algorithms on real-world prototypes. Equipped with a rigorously honed foundation in algorithmic optimization and a vehement zeal for innovation, I am confident in my readiness not only to contribute substantively to the department’s pioneering projects but also to translate academic erudition into pragmatic value for the future engineering community.